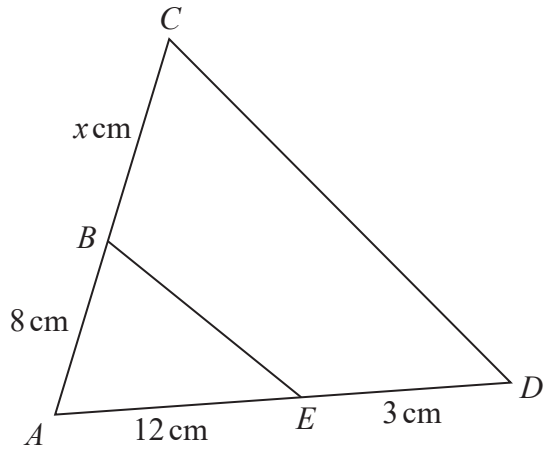
22 The two triangles in the diagram are similar.



There are two possible values of x.

Work out each of these values.
State any assumptions you make in your working.

AD = 12 + 3 = 15 cm; AC = x + 8 cm.

Case 1: assume AB corresponds to AC, and AE to AD.

	a	b
	8	12
$\times\frac{5}{4}$		$\times\frac{5}{4}$
	10	15

a: AB → AC; b: AE → AD.

Scale factor = 15 ÷ 12 = 5/4.

1 mark
Use matching corresponding sides

AC = 8 × 5/4 = 10, so x = 10 − 8 = 2 cm.

1 mark
First value: 2 cm

Case 2: assume AB corresponds to AD, and AE to AC.

	a	b
	8	12
$\times\frac{15}{8}$		$\times\frac{15}{8}$
	15	22.5

a: AB → AD; b: AE → AC.

Scale factor = 15 ÷ 8.

1 mark
Use the second possible correspondence

AC = 12 × 15/8 = 22.5, so x = 22.5 − 8 = 14.5 cm.

1 mark
Second value: 14.5 cm

1 mark
Both side correspondences stated

(Total for Question 22 is 5 marks)